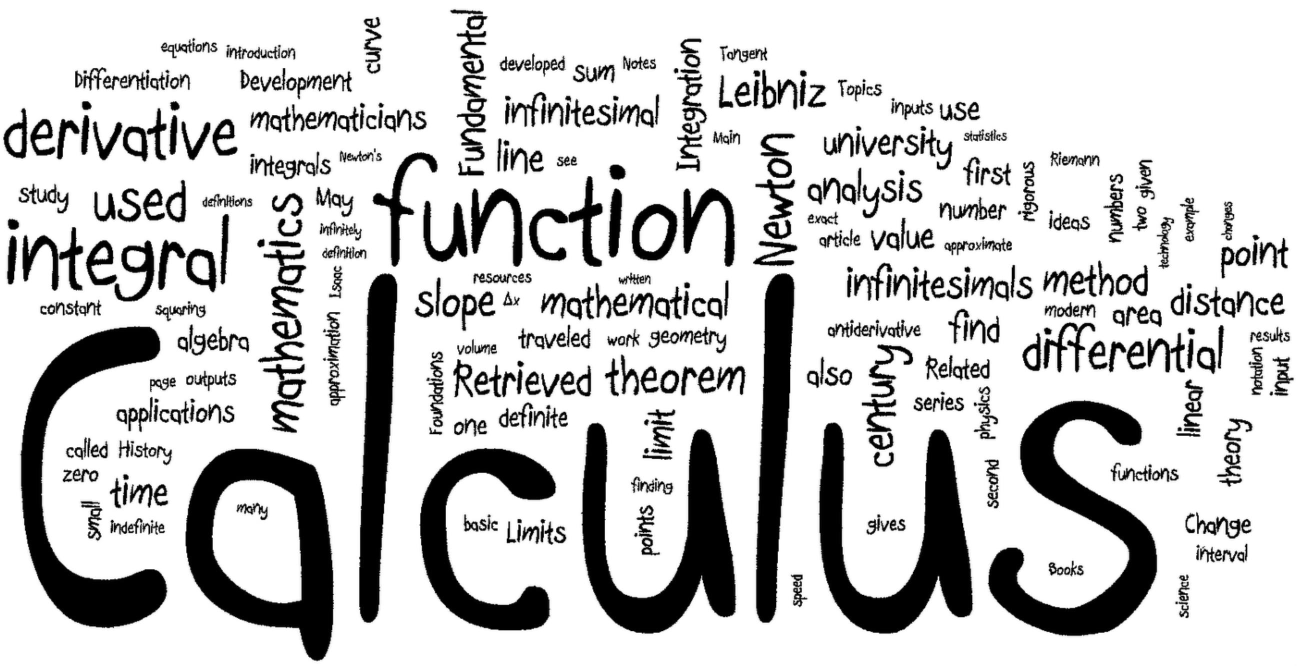
University of Zimbabwe

HMTH101 CALCULUS 1

Calculus of Single Variables

Author:
G. Nhawu

Department:
Mathematics



August 27, 2026

Contents

Chapter 3

Sequences

A girl and a boy jump into a river. The boy swims over to the girl and says, "God, it's cold." What's the probability they will kiss?

—Jenny Downham, You Against Me

Definition 3.0.1. *A* ***sequence*** *is a set of numbers $u_1, u_2, u_3, \ldots$ in a definite order of arrangement and formed according to a definite rule.*

Each number in the sequence is called a *term* and u_n is called the *nth term*. The sequence $u_1, u_2, u_3, \ldots$ is written briefly as $\{u_n\}$, e.g., $\{u_n\} = 2n$, where $u_1 = 2$, $u_2 = 4$, $u_3 = 6$ and so on. A sequence can be *finite* or *infinite*. A finite sequence has only definite number of terms. An infinite sequence is non-terminating, e.g., $u_n = \dfrac{1}{2n-1}$.

Recursive Definitions. The values of the terms in a sequence may be given explicitly by formulas such as $u_n = n^3 - 73n$.

A **recursive calculation** finds the value of u_n by looking to see which terms u_n depends on, then which terms depend, and so on.

Example: $u_0 = 1, \quad u_{n+1} = \dfrac{2}{u_n}, \quad n \in \mathbb{N}$.

Solution:

$$\begin{aligned} u_1 &= u_{0+1} = \frac{2}{u_0} = \frac{2}{1} = 2 \\ u_2 &= u_{1+1} = \frac{2}{u_1} = \frac{2}{2} = 1 \\ u_3 &= u_{2+1} = \frac{2}{u_2} = \frac{2}{1} = 2. \end{aligned}$$

You Try It: Define recursively

$$a_0 = a_1 = 1, \quad \text{and} \quad a_n = a_{n-1} + 2a_{n-2},\ n \geq 2.$$

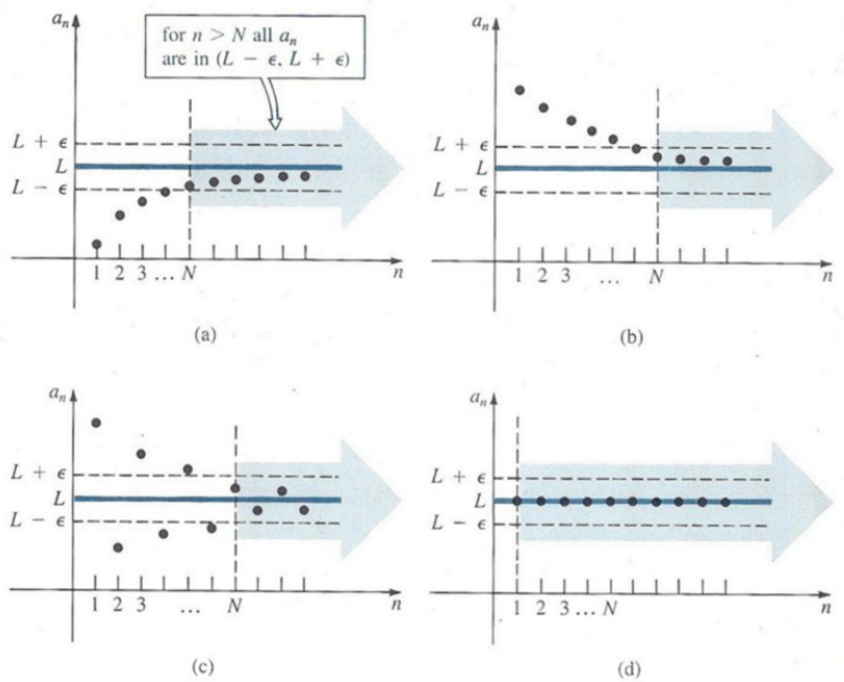
Find a_6 recursively.

3.1 Limits of Sequences

Lets consider the sequence $u_n = \dfrac{1}{n}$. The sequence has the terms $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$. We see that the terms of the sequence *tend to* or *approach* 0.

Definition 3.1.1. *A number L is called the* ***limit*** *of an infinite sequence $a_1, a_2, a_3, \ldots$ or $\{a_n\}$, if for any positive number ε, we can find a positive number N depending on ε such that $|a_n - L| < \varepsilon$ for all integers $n > N$. We write $\lim\limits_{n\to\infty} a_n = L$.*

If $\{a_n\}$ is a convergent sequence, it means that the terms a_n can be made arbitrarily close to L for n sufficiently large.



Example: If $u_n = 3 + \dfrac{1}{n} = \dfrac{3n+1}{n}$, the sequence is $4, \frac{7}{2}, \frac{10}{3}, \ldots$ and we can show that $\lim\limits_{n\to\infty} u_n = 3$.

If the limit of a sequence exists, the sequence is called **convergent**, otherwise, it is called **divergent**.

Example: Prove that $\lim_{n\to\infty} \frac{1}{n} = 0$.

Proof: Let $\varepsilon > 0$, we can find $N(\varepsilon)$ such that $\left|\frac{1}{n} - 0\right| = \left|\frac{1}{n}\right| = \frac{1}{n} < \varepsilon$. But $n > \frac{1}{\varepsilon}$. So $N = \frac{1}{\varepsilon}$. Taking N to be the smallest integer greater than $\frac{1}{\varepsilon}$, we have, $\lim_{n\to\infty} \frac{1}{n} = 0$.

You Try It: Prove that $\lim_{n\to 0} \frac{1}{n^p} = 0$ if $p \in \mathbb{N}$.

Example: Use the definition of a limit to prove that $\lim_{n\to\infty} \frac{2n-1}{3n+2} = \frac{2}{3}$.

Proof: Let $\varepsilon > 0$, we can find $N(\varepsilon)$ such that

$$\left|\frac{2n-1}{3n+2} - \frac{2}{3}\right| = \left|\frac{3(2n-1) - 2(3n+3)}{3(3n+2)}\right| = \left|\frac{6n-3-6n-4}{3(3n+2)}\right| = \left|\frac{-7}{3(3n+2)}\right| = \left|\frac{7}{3(3n+2)}\right| < \varepsilon$$

$$\begin{aligned} \frac{7}{3(3n+2)} &< \varepsilon \\ n &> \frac{7-6\varepsilon}{9\varepsilon}. \end{aligned}$$

Take $N = \frac{7-6\varepsilon}{9\varepsilon}$. So taking N to be the smallest integer greater than $\frac{7-6\varepsilon}{9\varepsilon}$, we have $\left|\frac{2n-1}{3n+2} - \frac{2}{3}\right| < \varepsilon$, i.e., $\lim_{n\to\infty} \frac{2n-1}{3n+2} = \frac{2}{3}$.

3.2 Theorems on Limits

If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, then

1. $\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = A + B$.
2. $\lim_{n\to\infty} (a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n = A - B$.
3. $\lim_{n\to\infty} (a_n \cdot b_n) = (\lim_{n\to\infty} a_n)(\lim_{n\to\infty} b_n) = AB$.
4. $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} = \frac{A}{B}$ if $\lim_{n\to\infty} b_n = B \neq 0$.
5. The limit of a convergent sequence $\{u_n\}$ of real numbers is unique.

Proof: We must show that if $\lim_{n\to\infty} u_n = l_1$ and $\lim_{n\to\infty} u_n = l_2$, then $l_1 = l_2$. By *hypothesis*, given any $\varepsilon > 0$, we can find N such that $|u_n - l_1| < \frac{\varepsilon}{2}$ when $n > N$ and $|u_n - l_2| < \frac{\varepsilon}{2}$ when $n > N$. Then

$$|l_1 - l_2| = |l_1 - u_n + u_n - l_2| \leq |l_1 - u_n| + |u_n - l_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

i.e., $|l_1 - l_2|$ is less than any positive ε (however small) and so must be zero, i.e., $l_1 - l_2 = 0 \Longrightarrow l_1 = l_2$.

Example: If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, prove that $\lim_{n\to\infty} (a_n + b_n) = A + B$.

Proof: We must show that for any $\varepsilon > 0$, we can find $N > 0$, such that $|(a_n + b_n) - (A + B)| < \varepsilon$ for all $n > N$. We have

$$|(a_n + b_n) - (A + B)| = |(a_n - A) + (b_n - B)| \leq |a_n - A| + |b_n - B|.$$

By *hypothesis*, given $\varepsilon > 0$ we can find N_1 and N_2 such that $|a_n - A| < \frac{\varepsilon}{2}$ for all $n > N_1$ and $|b_n - B| < \frac{\varepsilon}{2}$ for all $n > N_2$. Then

$$|(a_n + b_n) - (A + B)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

for all $n > N$ where $N = \max(N_1, N_2)$. Hence $\lim_{n\to\infty} (a_n + b_n) = A + B$.

3.3 Sequences Tending to Infinity

n tends to infinity, $n \to \infty$ (n grows or increases beyond any limit). Infinity is not a number and the sequences that tend to infinity are not convergent.

We write $\lim_{n\to\infty} a_n = \infty$, if for each positive number M, we can find a positive number N (depending on M) such that $a_n > M$ for all $n > N$.

Similarly, we write $\lim_{n\to\infty} a_n = -\infty$, if for each positive number M, we can find a positive number N such that $a_n < -M$ for all $n > N$.

Example: Prove that (a) $\lim_{n\to\infty} 3^{2n-1} = \infty$ (b) $\lim_{n\to\infty} (1 - 2n) = -\infty$.

Proof: (a) If for each positive number M we can find a positive number N such that $a_n > M$ for all $n > N$, then $3^{2n-1} > M$ when $(2n - 1)\ln 3 > \ln M$, i.e., $n > \frac{1}{2}\left(\frac{\ln M}{\ln 3} + 1\right)$. Taking N to be the smallest greater than $\frac{1}{2}\left(\frac{\ln M}{\ln 3} + 1\right)$, then $\lim_{n\to\infty} 3^{2n-1} = \infty$.

(b) If for each positive number M, we can find a positive number N such that $a_n < -M$ for all $n > N$, i.e., $1 - 2n < -M$ when $2n - 1 > M$ or $n > \frac{1}{2}(M+1)$. Taking N to be the smallest integer greater than $\frac{1}{2}(M+1)$, we have $\lim_{n\to\infty}(1-2n) = -\infty$.

3.4 Bounded and Monotonic Sequences

A sequence that tends to a limit l is said to be convergent and the sequence converges to l. A sequence may tend to $+\infty$ or $-\infty$, and is said to be divergent and it diverges to $+\infty$ or $-\infty$.

If $u_n \le M$ for $n = 1, 2, 3, \ldots$, where M is a constant, we say that the sequence $\{u_n\}$ is *bounded above* and M is called an *upper bound*. The smallest upper bound is called the *least upper bound* (l.u.b).

If $u_n \ge m$, the sequence is *bounded below* and m is called a *lower bound*. The largest lower bound is called the *greatest lower bound* (g.l.b).

If $m \le u_n \le M$, the sequence is called *bounded*, indicated by $|u_n| \le P$. (Every convergent sequence is bounded, but the converse is not necessarily true)

If $u_{n+1} \ge u_n$, the sequence is called *monotonic increasing* and if $u_{n+1} > u_n$ it is called *strictly increasing*. If $u_{n+1} \le u_n$, the sequence is called *monotonic decreasing*, while if $u_{n+1} < u_n$ it is *strictly decreasing*.

Examples: 1. The sequence $1, 1.1, 1.11, 1.111, \ldots$ is bounded and monotonic increasing.
2. The sequence $1, -1, 1, -1, 1, \ldots$ is bounded but not monotonic increasing or decreasing.

Definition 3.4.1. *A* ***null sequence*** *is a sequence that converges to* 0*, e.g.,* $u_n = \dfrac{1}{n-10}$, $n \ge 11$.

If $\{u_n\}$ does not tend to a limit or $+\infty$ or $-\infty$, we say that $\{u_n\}$ *oscillates* (or is an oscillating sequence). It can oscillate finitely (bounded) or infinitely (unbounded).

Examples: $u_n = (-1)^n, \quad u_n = (-1)^n n$.

3.5 Limits of Combination of Sequences

We want to be able to evaluate limits, for example, of the form $\lim_{n\to\infty}\left(2 - \dfrac{1}{n} + \dfrac{3}{n^2}\right)$ or $\lim_{n\to\infty}\dfrac{5-2n^2}{4+3n+2n^2}$.

Example: $\lim_{n\to\infty}\left(2-\frac{1}{n}+\frac{3}{n^2}\right) = \lim_{n\to\infty} 2 - \lim_{n\to\infty}\frac{1}{n} + 3\lim_{n\to\infty}\frac{1}{n^2} = 2-0+0=2.$

Example: $\lim_{n\to\infty}\frac{3n^2-5n}{5n^2+2n-6} = \lim_{n\to\infty}\frac{3-\frac{5}{n}}{5+\frac{2}{n}-\frac{6}{n^2}} = \frac{3+0}{5+0+0} = \frac{3}{5}.$

Example: $\lim_{n\to\infty}(\sqrt{n+1}-\sqrt{n}) = \lim_{n\to\infty}(\sqrt{n+1}-\sqrt{n})\cdot\left(\frac{\sqrt{n+1}+\sqrt{n}}{\sqrt{n+1}+\sqrt{n}}\right) = \lim_{n\to\infty}\frac{1}{\sqrt{n+1}+\sqrt{n}} = 0.$

3.6 Squeeze Theorem

If $\lim_{n\to\infty} a_n = l = \lim_{n\to\infty} b_n$ and there exists an N such that $a_n \le c_n \le b_n$, for all $n > N$, then $\lim_{n\to\infty} c_n = l$.

Example: Find $\lim_{n\to\infty}\frac{\cos n}{n}$.

Solution: We know that $-1 \le \cos n \le 1$
$\implies -\frac{1}{n} \le \frac{\cos n}{n} \le \frac{1}{n} \implies -\lim_{n\to\infty}\frac{1}{n} \le \lim_{n\to\infty}\frac{\cos n}{n} \le \lim_{n\to\infty}\frac{1}{n} \implies 0 \le \lim_{n\to\infty}\frac{\cos n}{n} \le 0$
$\implies \lim_{n\to\infty}\frac{\cos n}{n} = 0.$

3.7 Tutorial Questions

1. Write the first five terms of each of the following sequences.
   (i) $\left\{\frac{2n-1}{3n+2}\right\}$ (ii) $\left\{\frac{1-(-1)^n}{n^3}\right\}$ (iii) $\left\{\frac{(-1)^{n-1}}{2\cdot 4\cdot 6\cdots 2n}\right\}$ (iv) $\left\{\frac{(-1)^{n-1}x^{2n-1}}{(2n-1)!}\right\}$
   (v) $\left\{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots+\frac{1}{2^n}\right\}$ (vi) $\left\{\frac{\cos nx}{x^2+n^2}\right\}$ (vii) $\left\{\frac{\sqrt{n}}{n+1}\right\}$.

2. Determine the general term of each sequence.
   (i) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \ldots.$
   (ii) $\frac{1}{2}, \frac{1}{12}, \frac{1}{30}, \frac{1}{56}, \frac{1}{90}, \ldots.$
   (iii) $\frac{1}{5^3}, \frac{3}{5^5}, \frac{5}{5^7}, \frac{9}{5^{11}}, \ldots.$

3. (i) Recursively define $b_0 = b_1 = 1$ and $b_n = 2b_{n-1} + b_{n-2}, n \ge 2$. Calculate b_5.
   (ii) Recursively define $a_0 = 0, a_1 = 1, a_2 = 2$ and $a_n = a_{n-1} - a_{n-2} + a_{n-3}$ for $n \ge 3$. List the first five terms.

(iii) Recursively define $s_0 = 1, s_1 = -3$ and $s_n = 6s_{n-1} - 9s_{n-2}$ for $n \geq 2$, find s_5.

4. Find the least positive integer N such that $\left|\dfrac{(3n+2)}{(n-1)} - 3\right| < \varepsilon$ for all $n > N$ if
(i) $\varepsilon = 0.01$ (ii) $\varepsilon = 0.001$ (iii) $\varepsilon = 0.0001$.

5. Use the definition of limits to show that $\lim\limits_{n\to\infty} \dfrac{2n-1}{3n+4}$ **cannot** be $\dfrac{1}{2}$.

6. If $\lim\limits_{n\to\infty} a_n = A$ and $\lim\limits_{n\to\infty} b_n = B$, prove that $\lim\limits_{n\to\infty} (a_n \cdot b_n) = AB$ and $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{A}{B}$ if $\lim\limits_{n\to\infty} b_n = B \neq 0$.

7. Use the definition of a limit to verify each of the following limits.
(i) $\lim\limits_{n\to\infty} \dfrac{2n-1}{3n+2} = \dfrac{2}{3}$ (ii) $\lim\limits_{n\to\infty} \dfrac{4-2n}{3n+2} = -\dfrac{2}{3}$ (iii) $\lim\limits_{n\to\infty} \dfrac{\sin n}{n} = 0$
(iv) $\lim\limits_{n\to\infty} (5n-2) = \infty$ (v) $\lim\limits_{n\to\infty} \dfrac{c}{n^p} = 0$, where $c \neq 0, p > 0$
(vi) $\lim\limits_{n\to\infty} (1-2n) = -\infty$ (vii) $\lim\limits_{n\to\infty} a^n = 0$ where $0 < |a| < 1$.

8. Use the properties of limits to evaluate each of the following limits.
(i) $\lim\limits_{n\to\infty} \dfrac{4-2n-3n^2}{2n^2+n}$ (ii) $\lim\limits_{n\to\infty} \left(\dfrac{\sqrt{3n^2-5n+4}}{2n-7}\right)$ (iii) $\lim\limits_{n\to\infty} (\sqrt{n^2+n} - n)$
(iv) $\lim\limits_{n\to\infty} \left(\dfrac{n(n+2)}{n+1} - \dfrac{n^3}{n^2+1}\right)$ (v) $\lim\limits_{n\to\infty} (\sqrt{n+1} - n)$ (vi) $\lim\limits_{n\to\infty} \left(\dfrac{2n-3}{3n+7}\right)^4$
(vii) $\lim\limits_{n\to\infty} (\sqrt{4n^2+n+5} - 2n)$ (viii) $\lim\limits_{n\to\infty} \sqrt[3]{\dfrac{(3-\sqrt{n})(\sqrt{n}+2)}{8n-4}}$.

9. If $u_{n+1} = \frac{1}{2}(u_n + \frac{2}{u_n})$ where $u_1 > 0$, show that $\lim\limits_{n\to\infty} u_n = \sqrt{2}$.

10. (i) If a sequence $\{u_n\}$ is recursively defined by $u_{n+1} = \sqrt{u_n + 1}, u_1 = 1$, prove that $\lim\limits_{n\to\infty} u_n = \dfrac{1}{2}(1+\sqrt{5})$.
(ii) Show by induction that the sequence $\{u_n\}$ is bounded above by 2.
(iii) Show by induction that $\{u_n\}$ is monotonic increasing.

11. (i) Let $\{x_n\}$ be the sequence defined by $x_1 = 1, x_{n+1} = \sqrt{3x_n + 1}$ for all $n \geq 1$. Prove that $\{x_n\}$ is an increasing function for every n.
(ii) Show that the sequence has $\dfrac{7}{2}$ as an upper bound i.e. $x_n \leq \dfrac{7}{2}$ for all n.

12. Show that if $a_n \to l$ as $n \to \infty$, then $a_{n+1} = \dfrac{2}{3}a_n + \dfrac{2}{3a_n^2}$ converges to $\sqrt[3]{2}$.

13. Let $a_n = \sqrt{n+1} - \sqrt{n}$. Show that $a_n \to 0$ as $n \to \infty$.

14. A sequence $\{u_n\}$ is such that $u_{n+3} = 6u_{n+2} - 5u_{n+1}$ and $u_1 = 2, u_2 = 6$. Prove that $u_n = 5^{n-1} + 1$.

15. A sequence $\{a_n\}$ is such that $a_1 = 1, a_2 = 3$ and $a_{n+2} = 3a_{n+1} - 2a_n$. Prove that $a_n = 2^n - 1$.

16. Define a sequence $x_1, x_2, x_3, \ldots$ by $x_1 = 5, x_2 = 13$ and $x_{n+1} = 5x_n - 6x_{n-1}$. Prove by induction that $x_n = 2^n + 3^n$.

17. Define a sequence $\{x_n\}$ recursively by $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$.

    (i) Show that $\{x_n\}$ is a monotone increasing sequence.

    (ii) Show that the sequence is bounded above by 2.

    (iii) State why the sequence converges.

    (iv) Find the limit of the sequence.